Workshop 8: Stochastic processes & wealth dynamics

FIE463: Numerical Methods in Macroeconomics and Finance using Python

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See GitHub repository for notebooks and data:

https://github.com/richardfoltyn/FIE463-V26

Exercise 1: Wealth dynamics with AR(1) returns

Recall the household wealth dynamics we studied in the previous lecture, where assets $a_{i,t}$ evolved according to

$$a_{i,t+1} = R \cdot s \cdot a_{i,t} + y_{i,t+1}$$

and we assumed a fixed savings rate s, a fixed gross return R and some stochastic income process $y_{i,t}$.

In this exercise, we alter this setting to *stochastic returns* that follow an AR(1) process, so that the model of wealth dynamics is now given by

$$\begin{aligned} a_{i,t+1} &= R_{i,t+1} \cdot s \cdot a_{i,t} + y_{i,t+1} \\ \log R_{i,t+1} &= \mu_r + \rho_r \log R_{i,t} + \epsilon_{i,t+1}, \qquad \epsilon_{i,t+1} \overset{\text{iid}}{\sim} \mathcal{N}\left(0, \sigma_r^2\right) \\ \log y_{i,t+1} &\overset{\text{iid}}{\sim} \mathcal{N}\left(\mu_y, \sigma_y^2\right) \end{aligned}$$

where $R_{i,t+1}$ follows an AR(1) in logs and $y_{i,t+1}$ is IID log-normally distributed (we ignore serial correlation in income in this exercise).

Brief detour: Why would risky returns be correlated over time?

The finance literature has identified two mechanisms that can create persistence in returns: *type dependence* and *scale dependence*.

- *Type dependence:* This refers to the mechanism where investors persistently earn higher average returns or choose certain risk levels due to their inherent, stable characteristics, regardless of their total wealth. These enduring individual traits often include natural investment skill, financial sophistication, innate risk tolerance, or entrepreneurial talent.
- *Scale dependence:* This describes the phenomenon where having a larger amount of net worth directly causes an investor to earn higher average returns or adopt a riskier asset allocation. This advantage typically occurs because wealthier households benefit from economies of scale, gain access to better exclusive investment opportunities, or naturally exhibit decreasing relative risk aversion as their wealth grows.

References

- Bach, L., Calvet, L. E., & Sodini, P. (2020). Rich pickings? Risk, return, and skill in household wealth. *American Economic Review*, *110*(9), 2703-2747.
- Calvet, L. E., & Sodini, P. (2014). Twin picks: Disentangling the determinants of risk-taking in household portfolios. *The Journal of Finance*, *69*(2), 867-906.
- Fagereng, A., Guiso, L., Malacrino, D., & Pistaferri, L. (2020). Heterogeneity and persistence in returns to wealth. *Econometrica*, *88*(1), 115-170.

Parameters

To remain comparable with the scenarios discussed in the lecture, we set the following parameters:

Parameter	Description	Value
s	Savings rate	0.75
σ_y	Volatility of log labor income	0.1
μ_y	Mean of log labor income	$-\frac{1}{2}\sigma_y^2$
ρ_r	Autocorrelation of log returns	0.6
σ_r	Volatility of log return shocks	0.2
μ_r	Intercept of log returns	$(1-\rho_r)\log(1.1) - \frac{1}{2}\frac{\sigma_r^2}{1+\rho_r}$

The parameter μ_y is chosen so that average income in levels is one, $\mathbb{E}[y_{i,t}] = 1$, while μ_r is chosen so that average gross returns are 1.1 as in the lecture, i.e., $\mathbb{E}[R_{i,t}] = 1.1$.

The following code defines the parameters class for this problem:

[1]:
```
import numpy as np
from dataclasses import dataclass


@dataclass
class Parameters:
    """
    Parameters for model with stochastic returns.
    """

    s: float = 0.75          # Exogenous savings rate
    sigma_y: float = 0.1     # Standard deviation of log income
    mu_y: float = -(sigma_y**2.0) / 2.0  # Mean of log income
    rho_r: float = 0.6       # Persistence of log gross returns
    sigma_r: float = 0.2     # Standard deviation of log gross returns
    mu_r: float = (1 - rho_r) * np.log(1.1) - sigma_r**2 / 2 / ( 1 + rho_r )  # Mean of
 ↪log gross returns
```

[2]:
```
# Create an instance of the Parameters class
par = Parameters()
```

The following code verifies that the calibration of parameters yields the desired moments $E[y_{i,t}] = 1$ and $E[R_{i,t}] = 1.1$. For this we use the formulas for the mean and variance of log-normal variables,

$$\log X \overset{\text{iid}}{\sim} \mathcal{N}\left(\mu, \sigma^2\right) \quad \Longrightarrow \quad \begin{cases} \mathbb{E}[X] & = e^{\mu + \frac{1}{2}\sigma^2} \\ \text{Var}(X) & = \left(e^{\sigma^2} - 1\right) e^{2\mu + \sigma^2} \end{cases}$$

[3]:
```
# Mean of income in levels (from log-normal formula)
y_mean = np.exp(par.mu_y + par.sigma_y**2 / 2)
```

```
# From the formulas for mean & variance of AR(1) process:
# Unconditional mean of log gross returns
uncond_mean_log_R = par.mu_r / (1 - par.rho_r)
# Unconditional variance of log gross returns
uncond_var_log_R = par.sigma_r**2 / (1 - par.rho_r**2)

# Unconditional mean of gross returns (from log-normal formula)
R_mean = np.exp(uncond_mean_log_R + uncond_var_log_R/2)

print(f'Mean income:       {y_mean:.3f}')
print(f'Mean gross return: {R_mean:.3f}')
```

```
Mean income:       1.000
Mean gross return: 1.100
```

Lastly, we can use the variance formula above to get an idea of the conditional standard deviation of returns:

[4]:
```
# Use variance formula for log-normal random variables
R_var_cond = (np.exp(par.sigma_r**2) - 1) * np.exp(2*par.mu_r + par.sigma_r**2)
print(f'Conditional standard deviation of gross returns: {np.sqrt(R_var_cond):.3f}')
```

```
Conditional standard deviation of gross returns: 0.211
```

This value is slightly higher than the volatility of annual returns of the S&P500, but still within reasonable bounds.

Tasks

We are interested in simulating the wealth dynamics and computing the Gini coefficient using the same approach as we did in the lecture.

1. Simulate gross returns for 1,000,000 periods and compute the unconditional mean. Verify that this mean is close to the calibration target of 1.1.

   Hint: Use the `simulate_ar1()` function we wrote in the lecture for this purpose. This function is included in the module `workshop08_ex1.py` for convenience.

2. Write a function `simulate_wealth()` to simulate the wealth trajectories of a panel of households who face AR(1) returns. The function signature should look as follows:

```
def simulate_wealth(par: Parameters, a0, T, N, rng=None):
    """
    Simulate the evolution of wealth when returns are stochastic.

    Parameters
    ----------
    par : Parameters
    a0 : float
        Initial wealth.
    T : int
        Number of time periods to simulate.
    N : int
        Number of individuals to simulate.
    rng : numpy.random.Generator, optional
        A random number generator instance.

    Returns
    -------
    a_sim : numpy.ndarray
```

```
        A (T+1, N) array of simulated wealth paths.
    """
```

Set the initial value of $\log R_{i,t}$ to the unconditional mean $\frac{\mu_r}{1-\rho_r}$ for all households.

Hint: Use the wealth simulation routine from the lecture as a template and make the necessary changes.

3. Using an initial wealth of $a_0 = 1$ for all households, simulate $N = 20$ households for $T = 100$ periods. Plot the wealth trajectories for these households in a single graph and also include the average simulated wealth.

4. Simulate a larger panel of $N = 1,000,000$ households for $T = 200$ periods. Compute the cross-sectional mean and variance of wealth for each period t and plot these in a figure with two subplots (one for the mean, one for the variance).

   How do these plots compare to the scenarios (with IID and AR(1) income) discussed in the lecture?

5. From the previous plots you suspect that the Gini coefficient changes somewhat across periods. Compute the cross-sectional Gini coefficients for each of the last 100 periods of your simulation and plot these Ginis as a time series. Add a horizontal line indicating the average Gini coefficient.

   Hint: Use the `gini()` function from the `workshop08_ex1` module for this purpose.

[5]:
```
# Enable automatic reloading of external modules
%load_ext autoreload
%autoreload 2
```

Solution.

Part 1 — Simulate gross returns

We import the `simulate_ar1()` function to simulate log gross returns, which follow an AR(1) process. We then compute the unconditional mean of gross returns in levels.

[6]:
```
from workshop08_ex1 import simulate_ar1

# Number of periods
T = 1_000_000

# Initial value (does not matter for this exercise)
log_R0 = np.log(R_mean)

# Simulate log gross returns
log_R_sim = simulate_ar1(log_R0, par.mu_r, par.rho_r, par.sigma_r, T)

# Convert to gross returns in levels
R_sim = np.exp(log_R_sim)

# Compute simulated mean
R_mean_sim = np.mean(R_sim)

print(f'Mean returns: exact = {R_mean:.3f}, simulated = {R_mean_sim:.3f}')
```

```
Mean returns: exact = 1.100, simulated = 1.101
```

As you can see, the mean of the simulated values is quite close to the exact analytical value.

Part 2 — Wealth simulation function

The following function implements the wealth simulation with AR(1) log returns and IID income. The code proceeds in six main steps:

1. Draw all IID log income realizations for all N households and all T periods and store them in a $T \times N$ array.
2. Draw all AR(1) return shock realizations for all N households and all T periods and store them in a $T \times N$ array.
3. Assume that all individuals start with the same log return which corresponds to the unconditional mean of the AR(1).
4. Set the initial assets for all households to the given value a_0.
5. Use the AR(1) law of motion to simulate next-period returns given current returns.
6. Use the asset law of motion to simulate next-period assets.

[7]:
```
def simulate_wealth(par: Parameters, a0, T, N, rng=None):
    """
    Simulate the evolution of wealth when returns are stochastic.

    Parameters
    ----------
    par : Parameters
    a0 : float
        Initial wealth.
    T : int
        Number of time periods to simulate.
    N : int
        Number of individuals to simulate.
    rng : numpy.random.Generator, optional
        A random number generator instance.

    Returns
    -------
    a_sim : numpy.ndarray
        A (T+1, N) array of simulated wealth paths.
    """

    if rng is None:
        rng = np.random.default_rng(seed=1234)

    # Random draws of income
    log_y = rng.normal(loc=par.mu_y, scale=par.sigma_y, size=(T, N))
    # Innovations to log gross returns
    epsilon_R = rng.normal(loc=0, scale=par.sigma_r, size=(T, N))

    # Income in levels
    y = np.exp(log_y)

    # Set initial value for log returns:
    # Unconditional mean of log gross returns
    log_R_mean = par.mu_r / (1 - par.rho_r)
    # Simulated AR(1) returns (store only one period at a time)
    log_R_sim = np.full(N, fill_value=log_R_mean)

    # Create array to store the simulated wealth paths
    a_sim = np.zeros((T + 1, N))
    a_sim[0] = a0

    for t in range(T):
```

```
        # Savings out of beginning-of-period assets
        savings = par.s * a_sim[t]

        # Log gross returns next period
        log_R_sim = par.mu_r + par.rho_r * log_R_sim + epsilon_R[t]

        # Gross returns next period
        R_sim = np.exp(log_R_sim)

        # Next-period assets
        a_sim[t + 1] = R_sim * savings + y[t]

    return a_sim
```

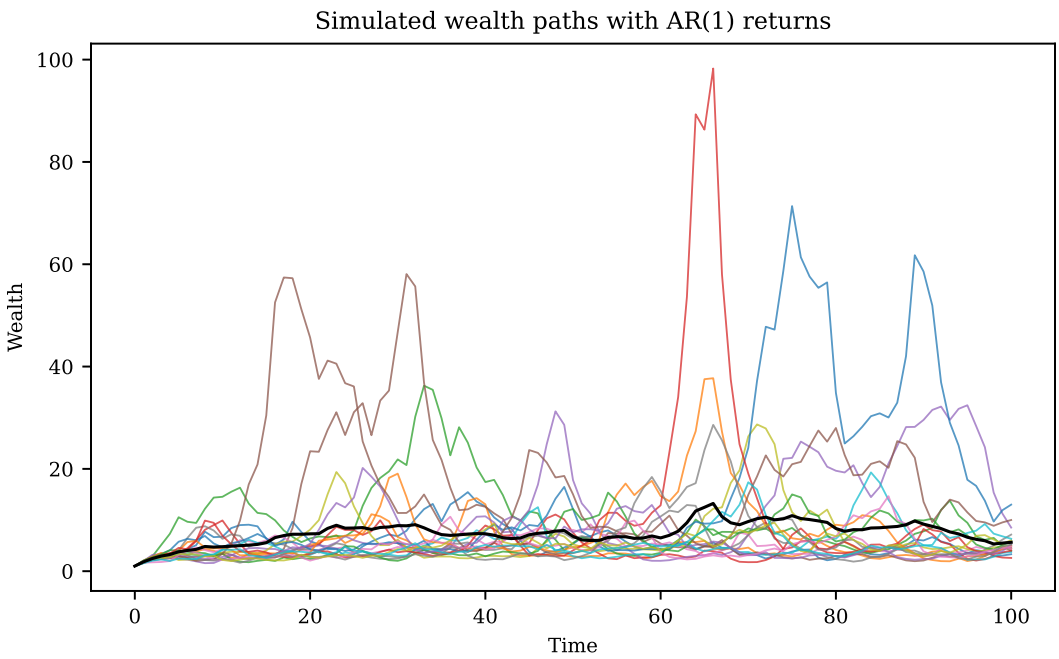
Part 3 — Simulate small panel

First, we simulate wealth trajectories for only a handful of households to get an idea of the dynamics.

[8]:
```
# Initial wealth (identical for all households)
a0 = 1.0
# Number of periods to simulate
T = 100
# Number of households to simulate
N = 20

# Simulate the wealth paths (result is an array of shape (T+1, N))
a_sim = simulate_wealth(par, a0, T, N)

# Mean of simulated time series
a_sim_mean = np.mean(a_sim, axis=1)
```

[9]:
```
import matplotlib.pyplot as plt

plt.figure(figsize=(7, 4))

plt.plot(a_sim, alpha=0.75, lw=0.75)
plt.xlabel('Time')
plt.ylabel('Wealth')
plt.title('Simulated wealth paths with AR(1) returns')
plt.plot(a_sim_mean, color='black', ls='-', lw=1.25)
```

[9]:
```
[<matplotlib.lines.Line2D at 0x7feecb028550>]
```



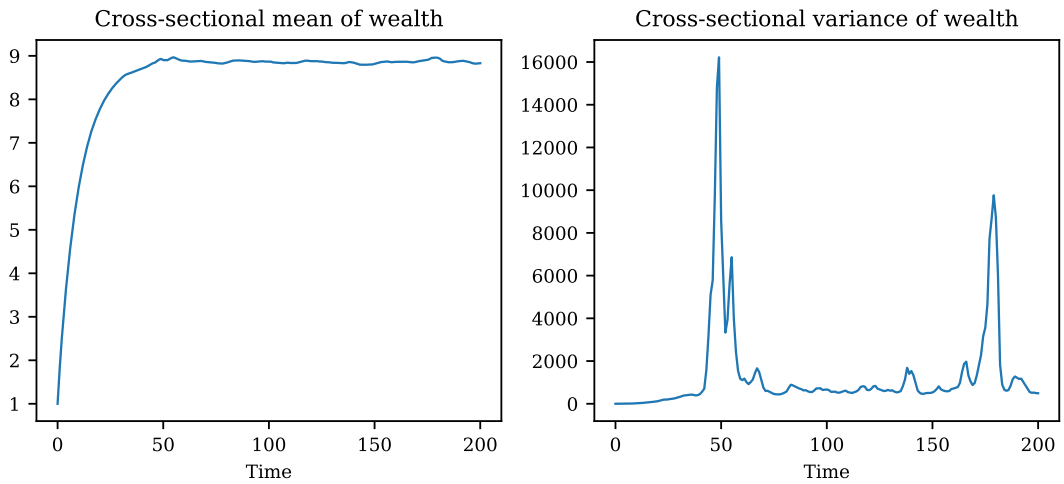
Part 4 — Simulate large panel

We now rerun the simulation for a much larger panel of households.

[10]:
```
# Simulate larger sample
N = 1_000_000
# Number of periods to simulate
T = 200

# Simulate the wealth paths (result is an array of shape (T+1, N))
a_sim = simulate_wealth(par, a0, T, N)

# Mean of simulated time series
a_sim_mean = np.mean(a_sim, axis=1)
a_sim_var = np.var(a_sim, axis=1)
```

[11]:
```
fig, (ax1, ax2) = plt.subplots(1, 2, figsize=(8, 3), sharex=True)

ax1.plot(a_sim_mean, lw=1)
ax1.set_xlabel('Time')
ax1.set_title('Cross-sectional mean of wealth')
ax2.plot(a_sim_var, lw=1)
ax2.set_title('Cross-sectional variance of wealth')
ax2.set_xlabel('Time')
```

[11]:
```
Text(0.5, 0, 'Time')
```



As the graphs indicate, the cross-sectional means and, in particular, variances are much jumpier than for the IID and AR(1) income scenarios studied in the lecture.

Part 5 — Gini coefficient

We loop over the last 100 periods of the simulation and compute the Gini coefficient for the cross section of wealth holdings in each period t.

[12]:
```
from workshop08_ex1 import gini

# Array to store Ginis for last 100 periods
ginis = np.empty(100)

# Iterate over the last 100 periods
for t in range(-100, 0):
    ginis[t] = gini(a_sim[t])
```

[13]:
```
# Plot cross-sectional Gini coefficients for last 100 periods
periods = np.arange(T-100, T)

plt.figure(figsize=(7, 4))
plt.plot(periods, ginis, color='steelblue')
plt.axhline(np.mean(ginis), color='black', lw=0.5, ls='--')
plt.xlabel('Period')
plt.title('Cross-sectional Gini coefficient')
```

[13]:
```
Text(0.5, 1.0, 'Cross-sectional Gini coefficient')
```

While the variances were quite volatile, the Gini does not change much across periods. As you can see, the Gini for this model is substantially higher than for the AR(1) income case discussed in the lecture, but still not at the level observed in the data (for example, the wealth Gini for the US is about 0.8). We could of course push up the Gini even further by increasing the persistence of returns, for example to $\rho_r = 0.7$.

Exercise 2: AR(1) vs Random Walk

Recall the AR(1) process we studied in the lecture, defined as

$$x_{i,t+1} = \rho x_{i,t} + \epsilon_{i,t+1}\,, \qquad \epsilon_{t+1} \overset{\text{iid}}{\sim} \mathcal{N}\left(0, \sigma^2\right)$$

In the lecture, we restricted our attention to the stationary case with $\rho \in (-1, 1)$. With $\rho = 1$, the above process is called a Random Walk, which is no longer stationary because its variance increases linearly over time.

To demonstrate this, perform the following tasks:

1. Write a function `simulate_panel()`, which simulates a *panel* of individuals i where each individual-specific realization $x_{i,t}$ follows the stochastic process defined above (assume that x_t represents an individual's log income). The function signature should look as follows:

```
def simulate_panel(rho, sigma, T, N, x0=0, rng=None):
    """
    Simulates a panel of stochastic processes.

    Parameters
    ----------
    rho : float
        The autoregressive parameter.
    sigma : float
        The standard deviation of the noise term.
    T : int
```

```
        The number of time periods to simulate.
    N : int
        The number of individuals to simulate.
    x0 : float, optional
        The initial value of the process.
    rng : Generator, optional
        Random number generator to use.

    Returns
    -------
    numpy.ndarray
        A (T+1, N) array with the simulated values.
    """
```

2. Let $\sigma = 0.1$. Simulate the trajectories of a cross section of $N = 100,000$ individuals for $T = 300$ periods for two different scenarios:
   1. AR(1) with $\rho = 0.9$;
   2. Random Walk with $\rho = 1$

   Make sure to use the same seed for both simulations.
3. Create a figure with two subplots:
   1. The first subplot should contain two lines showing the average value of the simulated AR(1) and Random Walk for each period t, i.e., average across N individuals for each t.
   2. The second subplot should contain two lines showing the variance of the simulated AR(1) and Random Walk for each period t.
4. Repeat the previous exercise, but use $\rho = 0.99$ for the AR(1) instead. How does the behavior of the cross-sectional mean and variance of the AR(1) change?

Solution.

Part 1 — Panel simulation function

The simulation routine is almost identical to `simulate_ar1()`, which we wrote earlier, but instead of a single value, we simulate the entire cross section of households one period at a time.

[14]:
```python
import numpy as np

def simulate_panel(rho, sigma, T, N, x0=0, rng=None):
    """
    Simulates a panel of stochastic processes.

    Parameters
    ----------
    rho : float
        The autoregressive parameter.
    sigma : float
        The standard deviation of the noise term.
    T : int
        The number of time periods to simulate.
    N : int
        The number of individuals to simulate.
    x0 : float, optional
        The initial value of the process.
    rng : Generator, optional
        Random number generator to use.
```

```
    Returns
    -------
    numpy.ndarray
        A (T+1, N) array with the simulated values.
    """

    # Create RNG instance
    if rng is None:
        rng = np.random.default_rng(seed=1234)

    # Create an array to store the simulated values
    x = np.zeros((T+1, N))

    # Set the initial value
    x[0] = x0

    # Draw random shocks
    eps = rng.normal(0, sigma, size=(T, N))

    # Simulate the stochastic process
    for t in range(T):
        x[t+1] = rho * x[t] + eps[t]

    return x
```

Part 2 — Simulate AR(1) and Random Walk

[15]:
```
# Standard deviation of innovations (both AR(1) and Random Walk)
sigma = 0.1
# Autocorrelation coefficient for AR(1) case
rho = 0.9

# Number of individuals
N = 100_000
# Number of periods
T = 300

# Seed for both simulations
seed = 1234

# Simulate the AR(1)
rng = np.random.default_rng(seed)
ar1_sim = simulate_panel(rho=rho, sigma=sigma, T=T, N=N, rng=rng)

# Simulate the Random Walk
rng = np.random.default_rng(seed)
rw_sim = simulate_panel(rho=1, sigma=sigma, T=T, N=N, rng=rng)
```

Part 3 — Plot mean and variance

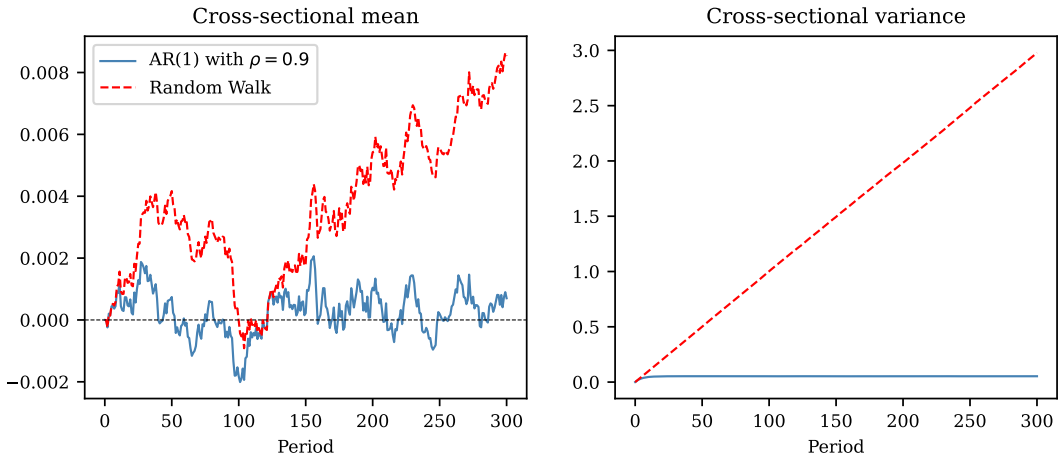
We first compute the mean and variance for both the AR(1) and the Random Walk.

[16]:
```
# Moments for AR(1)
ar1_sim_mean = np.mean(ar1_sim, axis=1)
ar1_sim_var = np.var(ar1_sim, axis=1)

# Moments for Random Walk
rw_sim_mean = np.mean(rw_sim, axis=1)
rw_sim_var = np.var(rw_sim, axis=1)
```

We can now plot these for each period:

[17]:
```
import matplotlib.pyplot as plt

fig, (ax0, ax1) = plt.subplots(1, 2, figsize=(8, 3), sharex=True)

ax0.plot(ar1_sim_mean, lw=1, color='steelblue', label=f'AR(1) with $\\rho={rho}$')
ax0.plot(rw_sim_mean, lw=1, color='red', ls='--', label='Random Walk')
ax0.set_xlabel('Period')
ax0.set_title('Cross-sectional mean')
ax0.axhline(0, color='black', lw=0.5, ls='--')
ax0.legend()

ax1.plot(ar1_sim_var, lw=1, color='steelblue', label='AR(1)')
ax1.plot(rw_sim_var, lw=1, color='red', ls='--', label='Random Walk')
ax1.set_xlabel('Period')
ax1.set_title('Cross-sectional variance')
```

[17]:
```
Text(0.5, 1.0, 'Cross-sectional variance')
```



As the figure shows, for the AR(1), the average simulated realizations remain close to their expected value of 0 and the cross-sectional variance quickly converges to its unconditional value $\sigma^2/(1-\rho^2)$.

Part 4 — High persistence AR(1)

We repeat the steps from Part 3 but simulate the AR(1) with $\rho = 0.99$ instead.

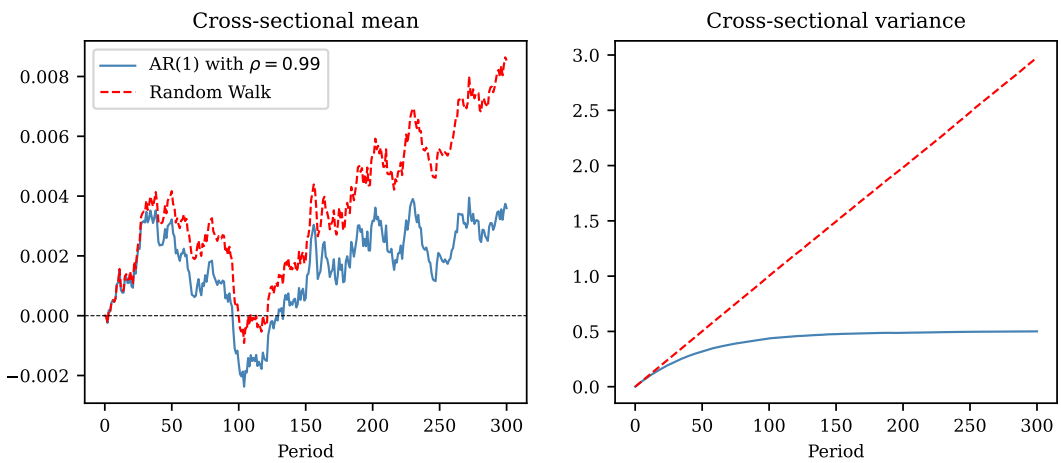
[18]:
```
# Standard deviation of innovations (both AR(1) and Random Walk)
sigma = 0.1
# Autocorrelation coefficient for AR(1) case
rho = 0.99

# Number of individuals
N = 100_000
# Number of periods
T = 300

# Seed for both simulations
seed = 1234

# Simulate the AR(1)
rng = np.random.default_rng(seed)
ar1_sim = simulate_panel(rho=rho, sigma=sigma, T=T, N=N, rng=rng)
```

```
# Simulate the Random Walk
rng = np.random.default_rng(seed)
rw_sim = simulate_panel(rho=1, sigma=sigma, T=T, N=N, rng=rng)
```

[19]:
```
# Moments for AR(1)
ar1_sim_mean = np.mean(ar1_sim, axis=1)
ar1_sim_var = np.var(ar1_sim, axis=1)

# Moments for Random Walk
rw_sim_mean = np.mean(rw_sim, axis=1)
rw_sim_var = np.var(rw_sim, axis=1)
```

[20]:
```
fig, (ax0, ax1) = plt.subplots(1, 2, figsize=(8, 3), sharex=True)

ax0.plot(ar1_sim_mean, lw=1, color='steelblue', label=f'AR(1) with $\\rho={rho}$')
ax0.plot(rw_sim_mean, lw=1, color='red', ls='--', label='Random Walk')
ax0.set_xlabel('Period')
ax0.set_title('Cross-sectional mean')
ax0.axhline(0, color='black', lw=0.5, ls='--')
ax0.legend()

ax1.plot(ar1_sim_var, lw=1, color='steelblue', label='AR(1)')
ax1.plot(rw_sim_var, lw=1, color='red', ls='--', label='Random Walk')
ax1.set_xlabel('Period')
ax1.set_title('Cross-sectional variance')
```

[20]:
```
Text(0.5, 1.0, 'Cross-sectional variance')
```



When the AR(1) features higher serial correlation (and thus less mean reversion), the cross-sectional mean deviates more strongly from its expected value, and the cross-sectional variance converges more slowly to its unconditional value.